Scaling Sharded Blockchains via GNN-based Partitioning and Batching

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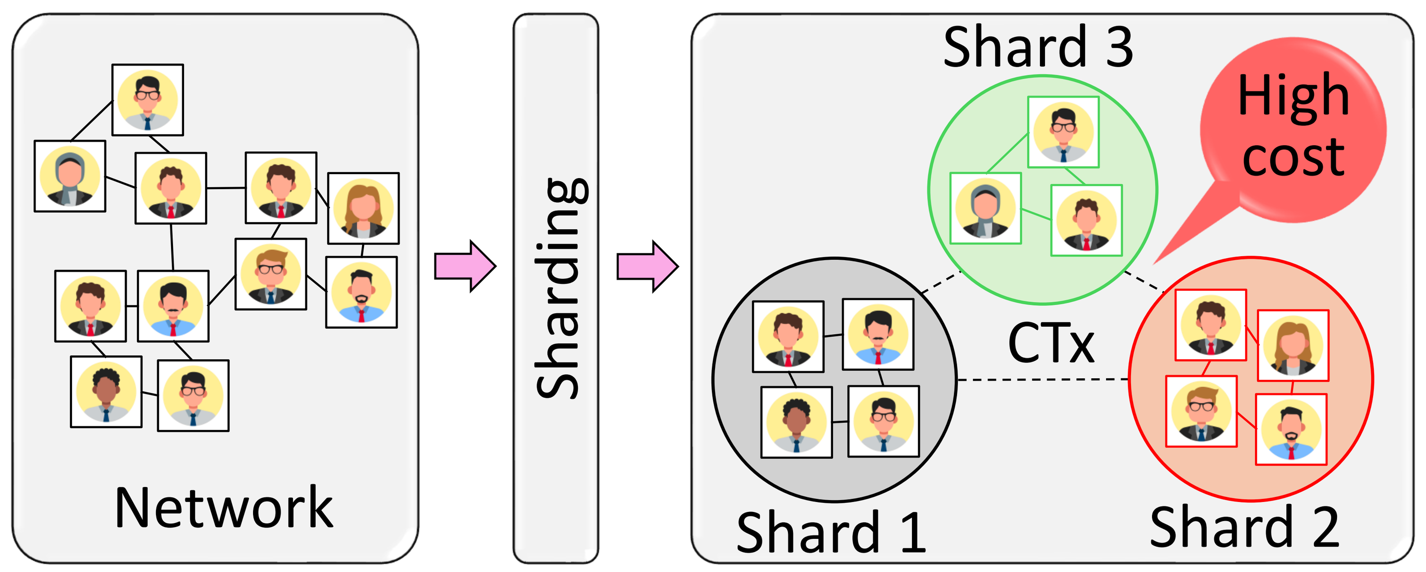
Introduction

- Blockchains struggle to scale, with throughput (TPS) far lower than centralized systems.

Bitcoin (~7 TPS) | Visa (~1,700 TPS)
Ethereum (~30 TPS) | MC (~2,000 TPS)

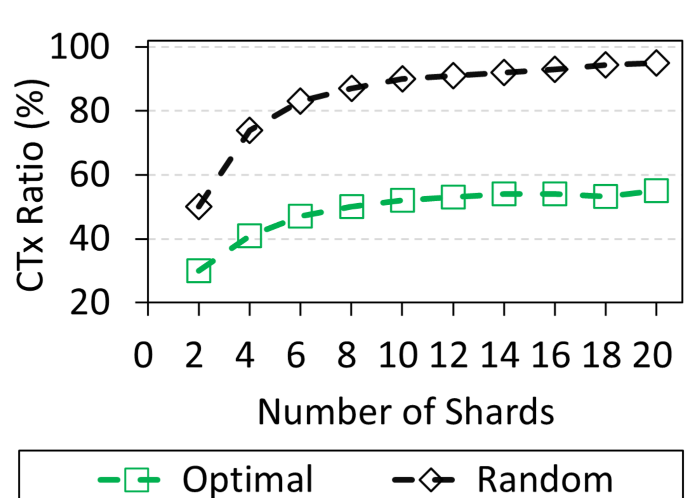
- Sharding is a popular scalability solution.



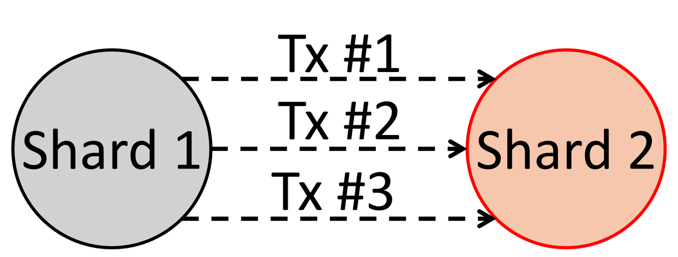
- Cross-shard tx (CTx) requires extra coordination, increasing overhead and limiting throughput.

Problem Statement

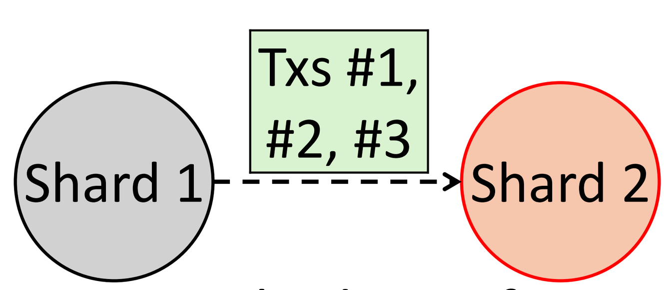
- **Research Question:** Can CTx be minimized?
- **Intuition:**
  - **H1:** Placing frequently interacting nodes within same shard reduces CTx.
  - **H2:** Caching and batched transfer of CTxs reduces communication cost.



full info of future tx | No historical tx considered



Individual transfer



Batched transfer

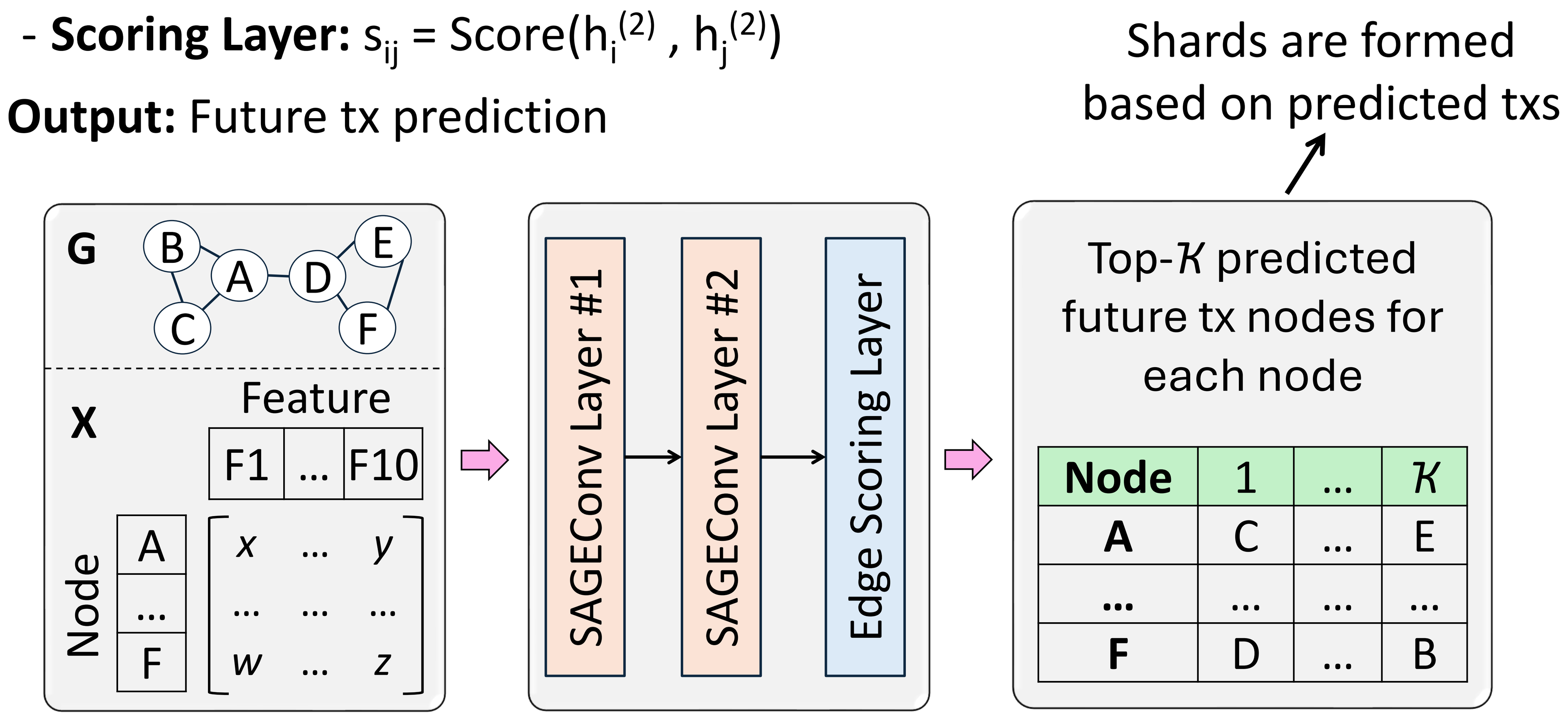
Research Methodology

Technique 1: GNN-Based Efficient Shard Formation (GSF)

- Predicting future txs is challenging due to the constantly evolving network.
- Graph Neural Networks (GNNs) improve shard assignment by learning patterns from the tx graph and node features.

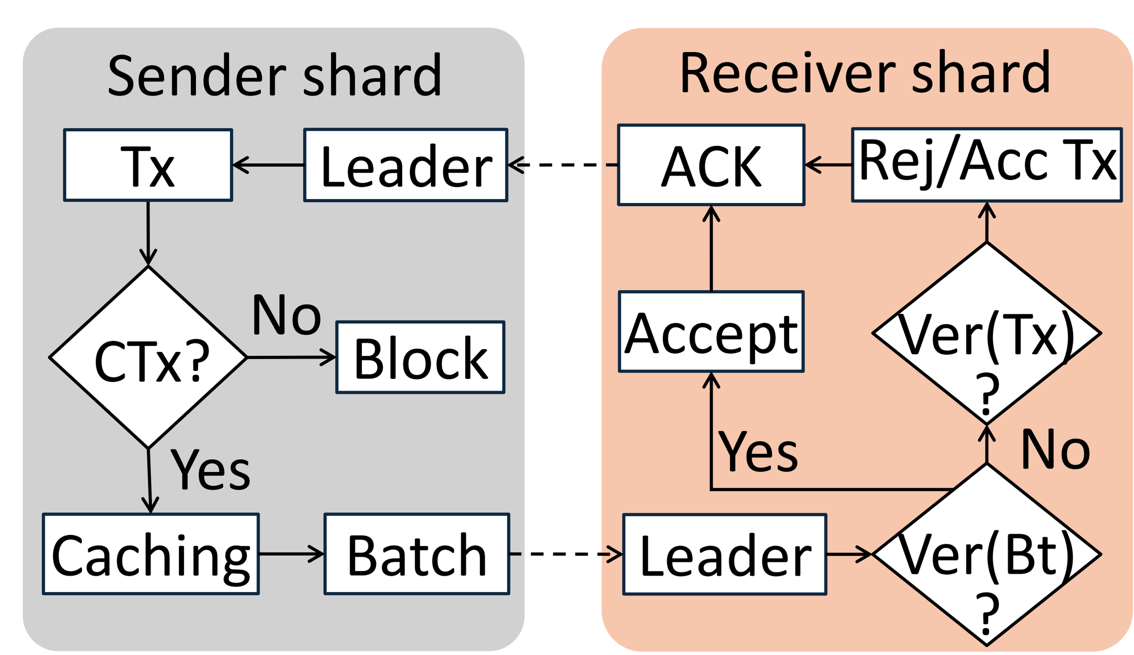
Model Design:

- **Input:** Transaction graph **G** and node feature matrix **X**
- **Architecture:** Two SAGEConv layers + Scoring layer
  - **Layer 1:** $H^{(1)} = \sigma(\mathrm{SAGEConv}_1(X, G))$, $\sigma(\cdot)$ ReLU activation function
  - **Layer 2:** $H^{(2)} = \mathrm{SAGEConv}_2(H^{(1)}, G)$
  - **Scoring Layer:** $s_{ij} = \mathrm{Score}(h_i^{(2)}, h_j^{(2)})$
- **Output:** Future tx prediction



Technique 2: Caching and Batched Cross-Shard Transfer (CBT)

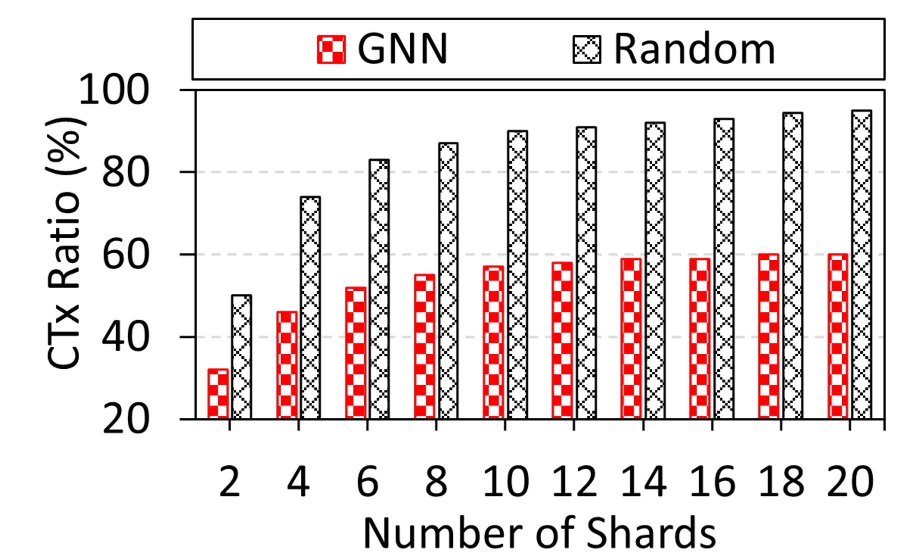
- Sender shard executes its portion of CTx and store in buffer table.
- With other CTxs, a batch is created.
- Batch is sent to receiver shard.
- Receiver shard verifies the batch and sends acknowledgement.



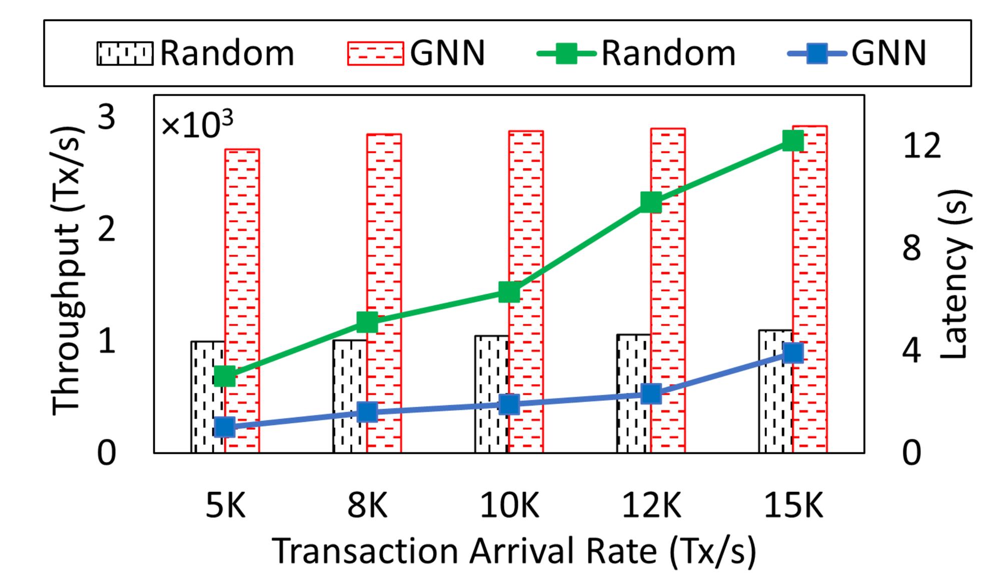
Evaluation Results

Result 1: GSF Accuracy and Performance

- GNN model accuracy around 80%.

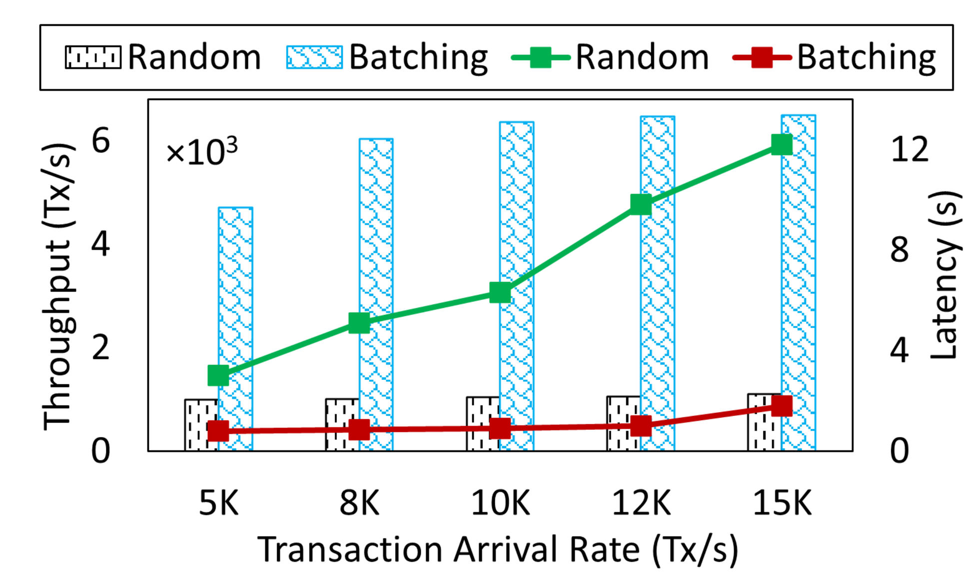


GNN partitioning reduces ~40% CTxs than Random



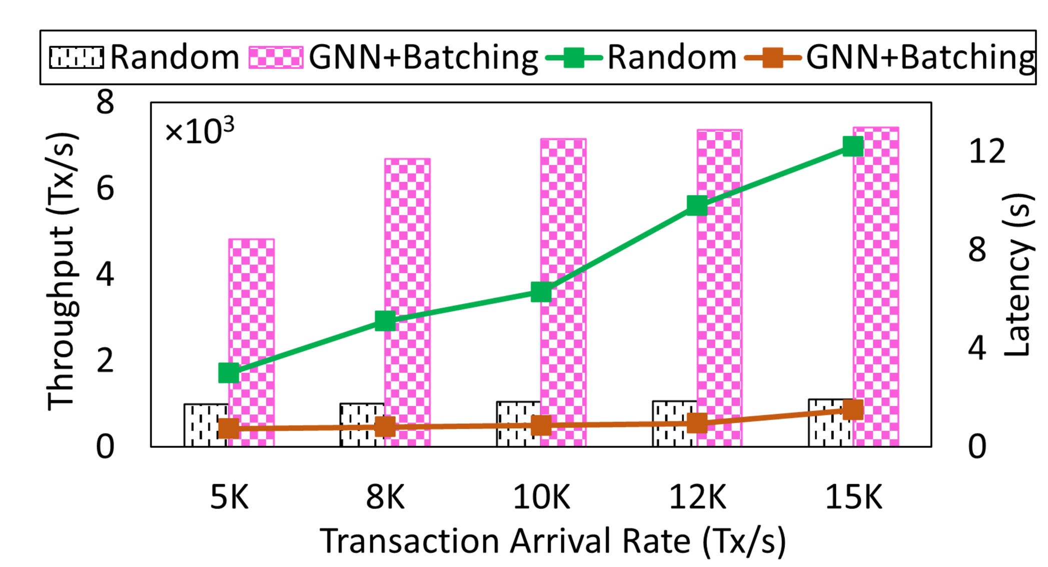
2.5× – 2.8× higher throughput and >65% lower latency with GNN

Result 2: CBT Performance



4.5× – 5.8× higher throughput and >70% lower latency with Batching

Result 3: Combined Performance of GSF and CBT



4.8× – 6.9× throughput gain and >75% latency reduction using both